

Outline	Content Area	Maple Commands
Arithmetic Computations Exact Arithmetic Integer Arithmetic Rational Manipulations Numerical Approximations Floating Point Arithmetic	Prime Numbers Fundamental Theorem of Arithmetic GCD, LCM Modular Arithmetic	isprime ifactor for gcd lcm evalf Digits numer denom normal
Algebraic Manipulations Multiplying Polynomials Factoring Polynomials Field Extension Simplifying Expressions Substitutions in Expressions	Binomial Theorem Pascal's Triangle Fundamental Theorem of Algebra	expand factor simplify subs binomial coeff coeffs sum
Solving Equations Equations in One Variable Exact Solutions Field Extension Approximate Solutions Field Extensions Algebraic Solutions Equations in Several Variables Linear Systems Non-Linear Systems	Solutions of Linear Systems Systems with Unique Solutions vs Systems with Infinitely Many Solutions vs Systems with No Solutions	solve fsolve

Graphing - Plots in Two Dimensions (in the Plane) - Explicit Functions of One Variable - Plot Options - List Plots (Scatter Plots) - Implicit Functions (Two Variables) - Parametric Plots - Plots in Three Dimensions (in Space) - Explicit Functions of Two Variables - Implicit Functions (Three Variables) - Contour Plots	Domain Range Intercepts Asymptotes - Vertical - Horizontal - Slant (Oblique) Monotonicity Local Extreme Values - Maxima - Minima	plot with(plots) implicitplot display plot3d implicitplot3d contourplot
Linear Algebra - Matrix Manipulations - Representing Matrices - Submatrix Extraction - Matrix Augmentation - Matrix Arithmetic - Addition - Scalar Multiplication - Multiplication - Inverses - Transpose - Determinants - Linear Systems of Equations - Augmented Matrix Representation - Elementary Row Operations - Gauss-Jordan Elimination - Linear Solution via Inverse - Cramer's Rule	See Outline for Content Topics	with(linalg) matrix evalm multiply submatrix augment inverse transpose determinant swaprow mulrow addrow backsub rref